

Notes: Card, DellaVigna, Funk & Iriberry (QJE, 2020)

Are Referees and Editors in Economics Gender Neutral?

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
2.1	Contribution relative to prior peer-review evidence	2
2.2	Contribution relative to studies that compare acceptance rates	3
2.3	Contribution relative to bias-by-gender-of-referee studies	3
2.4	Contribution relative to methodology and research transparency	3
3	Data and measurement	3
3.1	Journal-submission database	3
3.2	Author gender and author-team categories	4
3.3	Referee recommendations	4
3.4	Citations and citation transformation	4
3.5	Controls	5
4	Empirical specifications	5
4.1	Referee assignment model	5
4.2	Referee recommendation model	5
4.3	Citation conditional on referee evaluations	5
4.4	Editor R&R decision model	6
4.5	Desk rejection and delay models	6
5	Identification and interpretation	6
5.1	What can be identified	6
5.2	Interpretation of citation premium	6
5.3	Limits of interpretation	6
6	Tables and figures	8
6.1	Table I: beliefs of economists about gender and review	8
6.2	Table II and Figure I: summary statistics	8
6.3	Figure II: referee assignment	9
6.4	Figure III and Table III: referee evaluation by author and referee gender	10
6.5	Figure IV and Table IV: citations conditional on referee evaluations	11
6.6	Table V and Figure V: robustness and counterfactual R&R rates	13
6.7	Table VI and Figure VI: published papers and referee informativeness	14
6.8	Table VII and Table VIII: desk rejection and editorial delays	15
7	Short summary	17
8	Conclusion	17
8.1	Core conclusion	17
8.2	Economic intuition	17
8.3	Takeaway	18

1 Big picture

- **Question.** Do referees and editors in economics evaluate female-authored papers differently from male-authored papers?
- **Setting.** The paper studies nearly 30,000 submissions to four leading economics journals: the *Journal of the European Economic Association*, the *Quarterly Journal of Economics*, the *Review of Economics and Statistics*, and the *Review of Economic Studies*.
- **Core data advantage.** The authors observe paper characteristics, author gender, referee gender, referee recommendations, editorial decisions, author prior publications, field controls, and eventual citations, including for rejected papers.
- **Main referee-gender result.** Referee gender does not meaningfully affect the relative assessment of female-authored versus male-authored papers. Male referees do not appear to be systematically harsher toward female-authored papers than female referees.
- **Main standards result.** Conditional on referee recommendations and observables, female-authored papers receive about 25% more citations than comparable male-authored papers.
- **Interpretation of citation premium.** If citations proxy for paper quality, then female-authored papers need to be better to obtain the same referee recommendation or editorial outcome. This implies a higher bar for female authors.
- **Editor result.** Editors largely follow referees' recommendations. Because referee recommendations underweight the later citation performance of female-authored papers, editors' R&R decisions also under-admit female-authored papers relative to a citation-maximizing benchmark.
- **Counterfactual magnitude.** The paper estimates that a citation-maximizing rule would raise the R&R probability for female-authored NDR papers by about 1.7 percentage points, from roughly 14.2% to 15.7%.
- **Desk-rejection result.** At the desk-rejection stage, editors treat female authors more favorably than at the referee stage, but still impose a higher bar than the citation-maximizing benchmark would imply.
- **Null results.** The paper finds no meaningful gender differences in referee informativeness, in editors' weighting of female versus male referees, or in editorial delays for female-authored papers.
- **Takeaway.** The central issue is not that male referees discriminate relative to female referees. The concern is that the editorial/referee system as a whole appears to hold female-authored papers to a higher citation standard.

2 Incremental contribution of the paper

2.1 Contribution relative to prior peer-review evidence

- Prior work on gender in peer review often studies one journal, one conference, grant proposals, or acceptance rates only.
- This paper uses a much larger and richer journal-submission database, covering four top economics journals and including both accepted and rejected papers.
- The paper observes referee identities and recommendations, editor decisions, and later citations, making it possible to separate the behavior of referees from that of editors.

Interpretation: The incremental contribution is not just sample size. It is the ability to observe the sequence from author gender to referee assignment, referee recommendation, editorial decision, and ex post citation impact.

2.2 Contribution relative to studies that compare acceptance rates

- Acceptance or R&R rates alone cannot tell whether gender differences reflect quality differences, referee behavior, editor behavior, or citation patterns.
- The paper introduces citations as an ex post outcome to infer whether papers with the same review outcome are of comparable impact.
- This allows the authors to ask whether female-authored papers are held to a higher quality threshold.

Interpretation: The paper’s benchmark is not equality of raw acceptance rates. The benchmark is whether editorial decisions are aligned with later citation impact across author-gender groups.

2.3 Contribution relative to bias-by-gender-of-referee studies

- A common hypothesis is that male referees are more biased against female authors than female referees.
- The paper directly tests this using referee-paper-level data and interactions between author gender and referee gender.
- It finds little evidence that male referees evaluate female-authored papers more harshly than female referees do.

Interpretation: This is a key conceptual contribution: evidence of a higher bar for female-authored papers need not come from differential bias by male referees. It can arise from the collective review and editorial process.

2.4 Contribution relative to methodology and research transparency

- The analysis follows a pre-analysis plan registered before completion of data collection.
- The paper also elicits beliefs from economists about likely gender effects, letting the authors compare results to practitioner expectations.
- The paper tests robustness across Google Scholar citations, SSCI citations, published and rejected papers, desk rejection decisions, referee informativeness, and editorial delay.

Interpretation: The incremental methodological contribution is a transparent, multi-stage empirical audit of journal evaluation, not a single regression result.

3 Data and measurement

3.1 Journal-submission database

- The database includes submissions to four major economics journals.
- The paper tracks about 30,000 submissions, including desk-rejected papers and non-desk-rejected papers.
- The non-desk-rejected sample with at least two referees has 15,147 papers.
- For each paper, the authors observe journal, submission year, author names, author publication histories, referee identities, referee recommendations, editorial decisions, and later citations.
- The data were anonymized, and the authors did not access the text of papers, reports, or editorial letters.

Interpretation: The data allow the authors to study the full editorial pipeline while respecting confidentiality.

3.2 Author gender and author-team categories

- Author gender is assigned using first names and internet searches.
- The paper validates the gender assignment process and reports very low misclassification for published papers.
- Papers are divided into five author-gender categories:
 1. all-male teams;
 2. all-female teams;
 3. mixed-gender teams in which the most published coauthor is female;
 4. other mixed-gender teams;
 5. gender undetermined.
- In the non-desk-rejected sample, all-male teams account for about two-thirds of papers, all-female teams about 7%, mixed senior-female teams about 4%, and other mixed teams about 17%.

Interpretation: Splitting mixed-gender teams by senior author gender matters because the assignment and citation patterns differ across these groups.

3.3 Referee recommendations

Referee recommendations are reported in Editorial Express categories, ranging from definitely reject to accept. The paper often uses:

- an ordered index of recommendations;
- a positive recommendation indicator, defined as weak R&R or better;
- fractions of recommendations in each category at the paper level.

Interpretation: Recommendations are the primary input editors use for R&R decisions. The paper studies whether recommendation quality differs by author and referee gender.

3.4 Citations and citation transformation

The main ex post outcome is Google Scholar citations collected in mid-2015. The paper uses the inverse hyperbolic sine transformation:

$$\text{asinh}(\text{citations}) = \log \left(\text{citations} + \sqrt{\text{citations}^2 + 1} \right). \quad (1)$$

- This transformation is defined at zero citations.
- For large citation counts it behaves similarly to a logarithm.
- Robustness checks also use SSCI citations and alternative citation transformations.

Interpretation: Citations are used as an ex post proxy for paper impact or quality. The key question is whether papers with the same referee evaluation and author observables receive different citations by author gender.

3.5 Controls

The main controls include:

- journal-by-submission-year fixed effects;
- number of authors;
- author prior publications in 35 high-impact journals;
- broad field indicators based on JEL codes;
- share of female authors in a paper’s subfields;
- indicator for gender-related topics;
- referee publication records in referee-level regressions.

Interpretation: These controls address the fact that author gender is correlated with field, team size, and prior publication record.

4 Empirical specifications

4.1 Referee assignment model

At the referee-paper level, the paper estimates linear probability models of the form

$$FemaleReferee_{ipr} = \alpha + \beta AuthorGender_i + \Gamma X_i + \Lambda Z_r + \epsilon_{ipr}, \quad (2)$$

where X_i includes paper characteristics and Z_r includes referee publication controls.

Interpretation: This model tests whether editors assign female referees disproportionately to female-authored papers after conditioning on field, author publications, team size, and referee characteristics.

4.2 Referee recommendation model

The main referee-evaluation model is

$$Recommendation_{ipr} = \alpha + \beta AuthorGender_i + \delta FemaleReferee_r + \theta (AuthorGender_i \times FemaleReferee_r) + \Gamma X_i + \epsilon_{ipr}. \quad (3)$$

- *Recommendation* is either the ordered recommendation index or an indicator for a positive recommendation.
- Interactions test whether female referees treat female-authored papers differently from male referees.

Interpretation: The key null is no differential gender bias by referee gender. The paper largely fails to reject this null.

4.3 Citation conditional on referee evaluations

The core citation model is

$$\sinh(Citations_i) = \alpha + \beta AuthorGender_i + \sum_k \delta_k RecShare_{ik} + \Gamma X_i + \epsilon_i, \quad (4)$$

where $RecShare_{ik}$ is the fraction of referee recommendations in category k .

Interpretation: If female-authored papers have higher citations conditional on recommendations and controls, then they are achieving the same recommendations with higher ex post impact.

4.4 Editor R&R decision model

For non-desk-rejected papers, the editor’s R&R decision is modeled as

$$Pr(R\&R_i = 1) = F\left(\alpha + \beta AuthorGender_i + \sum_k \delta_k RecShare_{ik} + \Gamma X_i\right), \quad (5)$$

where F is a probit link in the paper’s main specifications.

Interpretation: Comparing this model with the citation model reveals whether editors put the same weight on author gender that a citation-maximizing decision rule would imply.

4.5 Desk rejection and delay models

The paper also studies desk rejection and editorial delay:

$$DecisionTime_i = \alpha + \beta AuthorGender_i + \Gamma X_i + \epsilon_i. \quad (6)$$

Interpretation: These analyses test whether gender differences arise before refereeing or through slower editorial treatment.

5 Identification and interpretation

5.1 What can be identified

- The paper can identify whether male and female referees treat author gender differently, conditional on observables.
- It can identify whether female-authored papers receive more citations conditional on referee recommendations and other observed characteristics.
- It can identify whether editors’ decisions align with citation-maximizing rules implied by the citation model.

Interpretation: The paper is not a randomized experiment in author gender. It is an observational audit using rich controls and ex post citation evidence.

5.2 Interpretation of citation premium

The citation premium for female-authored papers can be interpreted as evidence that female-authored papers face a higher bar if:

1. citations are a reasonable proxy for paper quality or impact;
2. controls adequately capture observably relevant differences;
3. citation practices are not biased in favor of female-authored papers by an offsetting amount.

Interpretation: Survey respondents expected female-authored papers to receive fewer citations, not more. Thus, if anything, citation bias would likely make the estimated higher-bar result conservative.

5.3 Limits of interpretation

- Citations are an imperfect measure of quality and may reflect field visibility, topic choice, and social networks.

- Author gender is not randomly assigned.
- The paper does not observe textual referee reports or papers themselves.
- The four journals are top journals and may not represent the entire economics profession.

Interpretation: The evidence is strongest for the narrow claim that the four-journal evaluation process does not appear citation-neutral across author gender.

6 Tables and figures

6.1 Table I: beliefs of economists about gender and review

Read:

- Survey respondents generally do not expect large gender differences in peer review.
- Editors are more likely than other economists to expect female-authored papers to be assigned to female referees.
- Respondents tend to expect that female-authored papers receive fewer citations than comparable male-authored papers, if there is any citation difference.

Economic message: The survey sets a benchmark: the paper’s finding that female-authored papers have higher citations conditional on review outcomes is not what economists generally expected.

TABLE I
SURVEY OF ECONOMISTS ABOUT THE ROLE OF AUTHOR AND REFEREE GENDER

Surveyed group:	All (1)	Editors (2)	Female economists (3)	Male economists (4)
Sample size				
Number surveyed	328	30	121	177
Number responded	141	14	60	67
Response rate	0.43	0.47	0.50	0.38
Classifying papers by author gender				
Consider an author team with both males and females, and the author with the most prior publications is female. Would you say that the patterns, in terms of the previous questions, would be more similar to:				
All-female	0.24	0.14	0.28	0.23
All-male	0.11	0.07	0.10	0.14
Halfway	0.41	0.50	0.33	0.47
It depends	0.23	0.29	0.28	0.17
If the author with the most prior publications is male, would you say that the patterns would be more similar to:				
All-female	0.01	0.00	0.00	0.02
All-male	0.56	0.64	0.68	0.44
Halfway	0.31	0.36	0.18	0.42
It depends	0.11	0.00	0.13	0.12
Referee assignment				
For two papers in the same field, are female-authored papers more likely to be assigned to female referees?				
More likely	0.19	0.36	0.23	0.11
Equally likely	0.77	0.64	0.70	0.87
Less likely	0.04	0.00	0.07	0.02
Referee assessment				
Consider the referee rec. for a female-authored paper with at least one male and at least one female referee.				
What percent of female referees are positive?	22.6	20.7	22.4	23.2
What percent of male referees are positive?	19.0	17.5	18.9	19.4
Consider the referee rec. for a male-authored paper with at least one male and at least one female referee.				
What percent of female referees are positive?	21.3	19.3	22.2	21.0
What percent of male referees are positive?	20.1	19.7	20.6	19.8
Editor assessment				
Holding constant the prior publication record of the author(s), the field of the paper, and the referee recs., do you think a female-authored paper has a higher, lower, or the same probability of receiving a R&R?				
More likely	0.11	0.14	0.03	0.18
About the same	0.67	0.57	0.63	0.72
Less likely	0.22	0.29	0.33	0.10
Citation discounting				
Conditional on field and quality, how large is the diff. in citations that a female-authored paper will receive?				
Mean citation gap in log points	-6.5	-3.8	-10.3	-3.9
Median citation gap in log points	0	0	-10	0

Notes. For legibility, questions are shortened from the original. Editor surveys were sent to the coeditors

6.2 Table II and Figure I: summary statistics

Read:

- Table II summarizes non-desk-rejected papers by author-gender category.
- All-female papers have lower average citations and lower R&R rates in raw data, but they also have fewer prior publications and different fields/team sizes.
- Mixed-gender senior-female teams have the highest raw citation counts and raw R&R rates.
- Figure I summarizes editorial decisions, referee recommendations, author publications, and referee publications by gender.

Economic message: Raw outcomes mix author gender with field, team size, and publication history. Later regressions condition on these differences.

TABLE II
SUMMARY STATISTICS FOR NON-DESK-REJECTED PAPERS

Sample:	All male (1)	All female (2)	Mix., F-led (3)	Mix., other (4)	Undet. (5)	All (6)
Google Scholar citations						
Asinh citations	2.72 (1.84)	2.61 (1.85)	3.09 (1.81)	2.92 (1.81)	2.15 (1.77)	2.74 (1.84)
Editorial decisions						
Received R&R decision	0.15	0.12	0.17	0.14	0.12	0.15
Author publications in 35 high-impact journals						
Publications: 0	0.32	0.59	0.00	0.22	0.53	0.32
Publications: 1	0.17	0.18	0.23	0.15	0.11	0.17
Publications: 2	0.12	0.09	0.22	0.13	0.11	0.12
Publications: 3	0.11	0.05	0.19	0.12	0.08	0.11
Publications: 4–5	0.13	0.07	0.20	0.16	0.08	0.13
Publications: 6+	0.16	0.02	0.17	0.22	0.09	0.15
Number of authors						
1 author	0.36	0.72	0.00	0.00	0.26	0.31
2 authors	0.43	0.25	0.53	0.46	0.39	0.42
3 authors	0.18	0.03	0.37	0.40	0.27	0.22
4+ authors	0.03	0.00	0.10	0.14	0.08	0.05
Field of paper						
Development	0.04	0.07	0.05	0.05	0.04	0.05
Econometrics	0.06	0.04	0.06	0.06	0.09	0.06
Finance	0.06	0.04	0.06	0.05	0.09	0.06
Health, urban, law	0.04	0.07	0.07	0.06	0.04	0.05
History	0.01	0.02	0.01	0.01	0.01	0.01
International	0.06	0.06	0.10	0.06	0.05	0.06
Industrial organization	0.05	0.05	0.04	0.05	0.05	0.05
Lab/experiments	0.02	0.02	0.04	0.04	0.02	0.03
Labor	0.11	0.20	0.14	0.15	0.10	0.12
Macro	0.11	0.06	0.08	0.08	0.10	0.10
Micro	0.11	0.08	0.09	0.11	0.09	0.11
Public	0.05	0.05	0.03	0.05	0.04	0.05
Theory	0.11	0.08	0.09	0.08	0.11	0.10
Unclassified	0.05	0.05	0.05	0.06	0.04	0.05
Missing field	0.11	0.12	0.10	0.11	0.13	0.11
Gender-field variables						
Share female in fields	0.15	0.18	0.17	0.17	0.15	0.16
Gender-topic fields	0.03	0.08	0.06	0.06	0.04	0.04
Referee recommendations						
Fraction definitely reject	0.12	0.12	0.10	0.12	0.19	0.12
Fraction reject	0.54	0.56	0.54	0.54	0.51	0.54
Fraction with no rec'n	0.06	0.06	0.04	0.05	0.06	0.06
Fraction weak R&R	0.10	0.10	0.12	0.11	0.10	0.10
Fraction R&R	0.10	0.10	0.12	0.10	0.09	0.10
Fraction strong R&R	0.04	0.04	0.04	0.04	0.03	0.04
Fraction accept	0.04	0.03	0.03	0.03	0.02	0.03

Table II, part 1

6.3 Figure II: referee assignment

Read:

- Papers by all-female authors are assigned to female referees at a higher rate than all-male papers.
- The raw all-female share of female referees is about 26%, compared with about 14% for all-male papers.
- After controls, female-authored papers are still around 7 percentage points more likely to receive a female referee.

Economic message: Editors appear to gender-match referees to female-authored papers, perhaps reflecting concerns about bias or field specialization.

TABLE II
(CONTINUED)

Sample:	All male (1)	All female (2)	Mix., F-led (3)	Mix., other (4)	Undet. (5)	All (6)
Referee publications in 35 high-impact journals						
Share of refs w/ 3+ publications	0.46	0.40	0.47	0.45	0.45	0.45
Referee gender (share per paper)						
Male	0.85	0.75	0.79	0.81	0.84	0.83
Female	0.14	0.24	0.20	0.18	0.14	0.15
Ambiguous	0.01	0.01	0.01	0.01	0.02	0.01
Number of observations	10,199	1,097	585	2,612	654	15,147
Share of papers	0.67	0.07	0.04	0.17	0.04	1.00

Table II, continuation

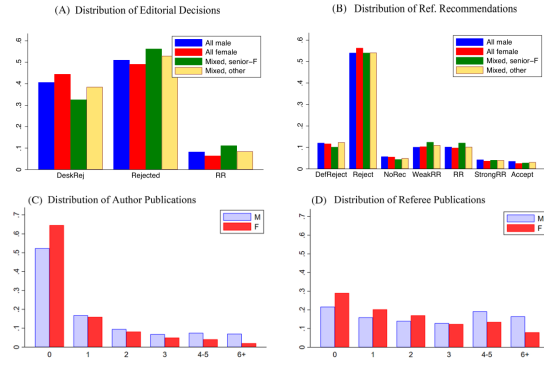


FIGURE I

Summary Statistics by Gender

This figure displays a few key summary statistics by gender. Panel A plots the distribution of the editor's decision, and Panel B shows the distribution of referee recommendations. Panel C plots the distribution of author publications in 35 high-impact journals in the five years leading up to submission, for the papers in our data set. Panel D reports the distribution of publications among referees by gender.

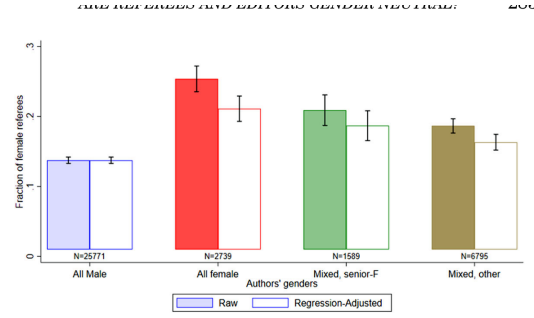


FIGURE II

Referee Assignment: Referee Gender as Function of Author Gender

Observations are at the referee-paper level. In the regression-adjusted bars, we adjust using the results in [Online Appendix Table 4, column \(3\)](#).

6.4 Figure III and Table III: referee evaluation by author and referee gender

Read:

- Figure III compares referee evaluations across author-gender groups and referee gender.
- The differences between male and female referees' treatment of female-authored papers are small.
- Table III formalizes this result using ordered recommendation indexes and positive recommendation indicators.
- Author-gender-by-referee-gender interactions are generally small and statistically insignificant.

Economic message: The data do not support a simple story in which male referees are systematically harsher toward female-authored papers than female referees are.

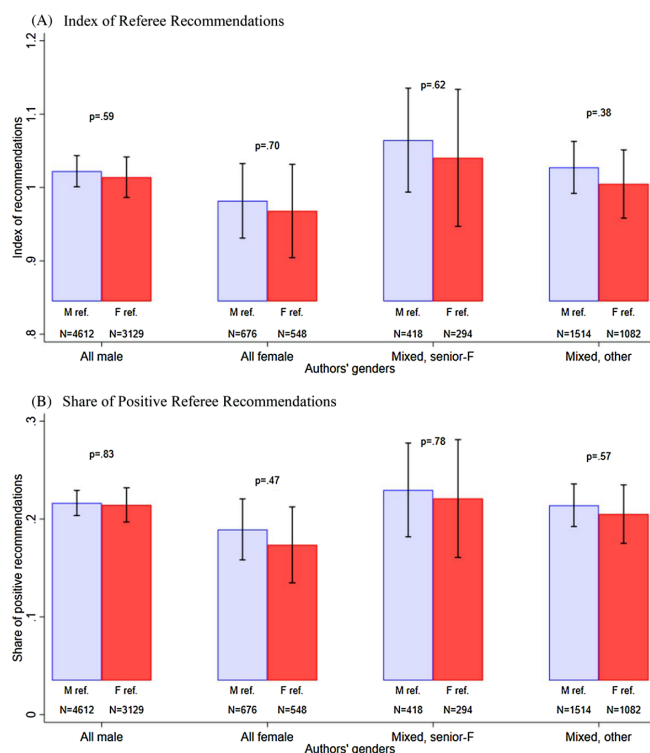


FIGURE III

Referee Evaluation by Author Gender and Referee Gender

Panel A displays the mean recommendation given by referees based on gender. The index of referee recommendations is constructed using the coefficients in the cites model in [Card and DellaVigna \(forthcoming\)](#). From Definitely Reject to Accept, the values are 0, 0.67, 1.01, 1.47, 1.92, 2.27, 2.33. The bands show two standard error intervals, clustered at the paper level. The sample includes only 4,341 papers with both male and female referees. Panel B shows the share of positive recommendations, defined as revise and resubmit or better. In both

TABLE III
REFEREE RECOMMENDATIONS, IMPACT OF AUTHOR TEAM GENDER

	OLS models for index of referee recommendations				Linear probability models for R&R recommendation or better			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
s (Omitted: all male authors)								
hors	-0.043 (0.014)	0.018 (0.014)			-0.029 (0.008)	0.001 (0.008)		
author team	0.045 (0.020)	0.009 (0.020)			0.014 (0.012)	-0.002 (0.012)		
or female								
author team	-0.015 (0.010)	-0.030 (0.011)			-0.011 (0.006)	-0.018 (0.006)		
l gender team	-0.121 (0.019)	-0.073 (0.019)			-0.049 (0.010)	-0.026 (0.010)		
Omitted: male referee)								
e			-0.025 (0.018)	—			-0.012 (0.012)	—
ions								
h. × Female ref.			0.009 (0.047)	0.004 (0.049)			-0.002 (0.030)	-0.007 (0.031)

Table III, part 1

TABLE III
(CONTINUED)

fiction:	OLS models for index of referee recommendations				Linear probability models for R&R recommendation or better			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ed auth. (F-senior) × Female ref.			-0.027 (0.062)	-0.024 (0.063)			-0.009 (0.041)	-0.007 (0.042)
ed auth. (other) × Female ref.			-0.009 (0.036)	-0.001 (0.037)			-0.003 (0.023)	0.001 (0.024)
determined auth. × Female ref.			0.055 (0.070)	0.048 (0.073)			0.004 (0.042)	-0.001 (0.044)
r fixed effects	No	No	Yes	Yes	No	No	Yes	Yes
ols for author pub., no. of authors, d, and gender-field	No	Yes	—	—	No	Yes	—	—
ols for journal-year	Yes	Yes	—	—	Yes	Yes	—	—
vl for referee pub.	No	No	Yes	Yes	No	No	Yes	Yes
vl for referee pub. × Author gender	No	No	No	Yes	No	No	No	Yes
vl for referee gender × Author pub	No	No	No	Yes	No	No	No	Yes
vl for referee gender × field	No	No	No	Yes	No	No	No	Yes
	38,840	38,840	38,840	38,840	38,840	38,840	38,840	38,840
	0.016	0.044	0.504	0.504	0.012	0.030	0.452	0.453

s. The index of referee recommendations is constructed using the coefficients in the cites model in [Card and DellaVigna \(forthcoming\)](#). From Definitely Reject to Accept, the are 0, 0.67, 1.01, 1.47, 1.92, 2.27, 2.33. Columns (3), (4), (7), and (8) also include a control for unknown-gender referee (coefficient not shown). In columns (4) and (8), Referee × Field includes gender-field variables.

Table III, continuation

6.5 Figure IV and Table IV: citations conditional on referee evaluations

Read:

- Figure IV shows that at nearly every referee-recommendation category, female-authored papers receive more residualized citations than male-authored papers.
- The pattern is similar for male and female referees.

- Table IV shows that controlling for recommendations, author publications, author team size, field, and journal-year, all-female papers have about a 0.22–0.24 higher $\text{asinh}(citationoutcome)$.
- In the R&R decision models, the corresponding author-gender coefficient is close to zero once controls are included.

Economic message: Female-authored papers achieve similar editorial outcomes despite having higher subsequent citations. This is the central higher-bar result.

ARE REFEREES AND EDITORS GENDER NEUTRAL? 295

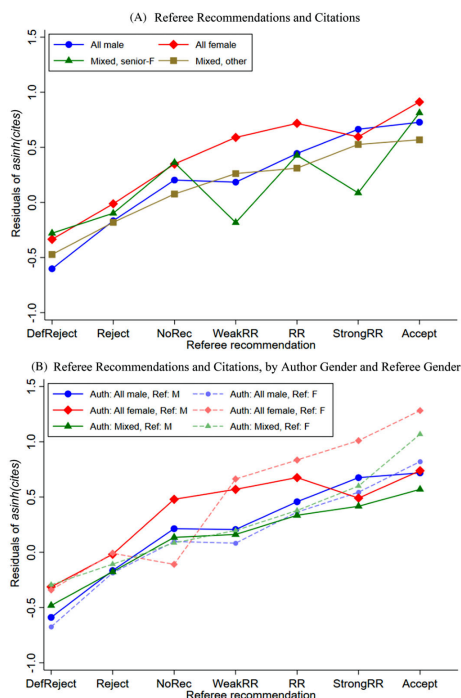


FIGURE IV

Differences in (Residualized) Citations and R&R Rate by Author Gender
Panels A and B show the weighted residuals of $\text{asinh}(citations)$ for a paper receiving a given recommendation, and Panel C shows the residuals of $\text{pr}(R\&R)$ for

Figure IV, panels A–B

296

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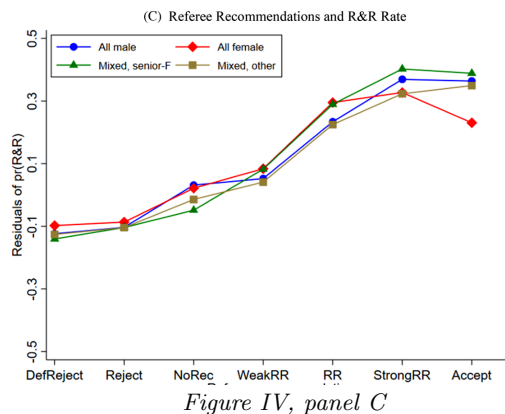


Figure IV, panel C

TABLE IV
CITATIONS AND EDITOR DECISIONS, IMPACT OF AUTHOR TEAM GENDER

Specification:	OLS models for <i>asinh</i> of Google Scholar citations				Probit models for receiving revise-and-resubmit		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Authors' genders							
All female	-0.07 (0.05)	0.24 (0.06)	0.22 (0.05)	0.22 (0.05)	-0.08 (0.06)	0.03 (0.05)	0.01 (0.06)
Mixed-gender author team	0.37 (0.08)	0.05 (0.07)	0.06 (0.07)	0.06 (0.07)	0.13 (0.07)	0.05 (0.07)	0.10 (0.07)
senior author female							
Mixed, other	0.26 (0.04)	-0.02 (0.04)	0.01 (0.04)	0.01 (0.04)	0.03 (0.05)	-0.08 (0.04)	-0.02 (0.05)
Undetermined	-0.36 (0.05)	-0.39 (0.06)	-0.31 (0.05)	-0.31 (0.05)	-0.09 (0.08)	-0.10 (0.06)	-0.06 (0.08)
Fractions of referee recommendations							
Reject	0.80 (0.06)		0.64 (0.06)	0.64 (0.06)	0.87 (0.15)		0.86 (0.16)
No recommendation	1.24 (0.12)		1.00 (0.10)	0.98 (0.10)	2.78 (0.17)		2.73 (0.18)
Weak R&R	1.73 (0.10)		1.47 (0.09)	1.45 (0.10)	3.16 (0.17)		3.16 (0.18)
R&R	2.33 (0.10)		1.93 (0.09)	1.89 (0.13)	4.63 (0.20)		4.61 (0.21)
Strong R&R	2.72 (0.15)		2.32 (0.13)	2.26 (0.22)	5.57 (0.21)		5.56 (0.21)
Accept	2.74 (0.13)		2.36 (0.12)	2.30 (0.19)	5.39 (0.21)		5.37 (0.22)
Author publications in 35 high-impact journals							
1 Publication		0.41 (0.04)	0.29 (0.04)	0.29 (0.04)		0.24 (0.04)	0.04 (0.05)
2 Publications		0.65 (0.04)	0.49 (0.04)	0.49 (0.04)		0.37 (0.05)	0.19 (0.07)
3 Publications		0.79 (0.04)	0.58 (0.04)	0.58 (0.04)		0.47 (0.05)	0.17 (0.07)
4-5 Publications		1.06 (0.06)	0.81 (0.06)	0.80 (0.06)		0.64 (0.05)	0.33 (0.06)
6+ Publications		1.31 (0.05)	0.99 (0.05)	0.99 (0.05)		0.82 (0.06)	0.41 (0.08)
Gender-field controls							
Share female in subfields		-0.03 (0.28)	-0.02 (0.26)	-0.01 (0.25)		-0.48 (0.28)	-0.35 (0.32)
Fraction of gender-topic subfields		-0.01 (0.09)	0.02 (0.10)	0.02 (0.10)		-0.09 (0.10)	-0.14 (0.14)
R&R indicator (mechanical publ. effect)				0.06 (0.14)			
Control function for selection (Value added of the editor)				0.32 (0.08)			
Editor leave-out-mean R&R Rate							3.43 (0.73)
Controls for no. of authors & field	No	Yes	Yes	Yes	No	Yes	Yes
Indicators for journal-year	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	15,147	15,147	15,147	15,147	15,147	15,147	15,147
<i>R</i> ² / pseudo <i>R</i> ²	0.20	0.20	0.26	0.27	0.48	0.07	0.49

6.6 Table V and Figure V: robustness and counterfactual R&R rates

Read:

- Table V tests robustness across citations, probit R&R models, publication status, sample years, papers with citations, and referee gender shares.
- The citation premium for all-female papers is robust across many specifications.
- Figure V translates the citation results into counterfactual R&R rates under citation-maximizing decision rules.
- The weighted average R&R rate for female-authored papers would rise from 14.2% to about 15.7% under the counterfactual rule.

Economic message: The citation premium is robust, and it implies a measurable under-admission of female-authored papers relative to a citation-maximizing benchmark.

CITATIONS AND EDITOR DECISIONS, ROBUSTNESS

	Coefficients in model 1			Coefficients in model 2			R&R rate for papers with at least 1 female author		
	AS, female		Mixed, female	AS, female		Mixed, female	Dis, female	Clin, female	pub
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
system dimension:									
heurak	0.22	0.08	0.01	0.01	0.10	0.02	0.142	0.110	0.157
β	(0.05)	(0.02)	(0.04)	(0.07)	(0.04)	(0.05)	(0.03)	(0.03)	(0.04)
task requirements:									
heurak	0.17	0.04	0.00	0.01	0.13	0.01	0.142	0.155	0.155
β	(0.05)	(0.02)	(0.04)	(0.07)	(0.02)	(0.05)	(0.03)	(0.03)	(0.04)
individual performance:									
heurak	0.05	0.07	0.01	0.01	0.07	0.02	0.142	0.155	0.154
β	(0.06)	(0.02)	(0.04)	(0.07)	(0.02)	(0.05)	(0.03)	(0.03)	(0.04)
task and individual performance:									
heurak	0.05	0.07	0.01	0.01	0.07	0.02	0.142	0.155	0.154
β	(0.06)	(0.02)	(0.04)	(0.07)	(0.02)	(0.05)	(0.03)	(0.03)	(0.04)
motivational measures:									
heurak	0.15	0.10	0.00	0.01	0.19	0.02	0.142	0.155	0.154
β	(0.05)	(0.03)	(0.03)	(0.06)	(0.02)	(0.05)	(0.03)	(0.03)	(0.04)
β	0.05	0.03	0.00	0.01	0.05	0.02	0.142	0.155	0.179
β	(0.04)	(0.02)	(0.03)	(0.07)	(0.06)	(0.05)	(0.03)	(0.03)	(0.04)
β	0.05	0.03	0.00	0.01	0.05	0.02	0.142	0.155	0.179
β	(0.04)	(0.02)	(0.03)	(0.07)	(0.06)	(0.05)	(0.03)	(0.03)	(0.04)
β	0.05	0.03	0.00	0.01	0.05	0.02	0.142	0.155	0.179
β	(0.04)	(0.02)	(0.03)	(0.07)	(0.06)	(0.05)	(0.03)	(0.03)	(0.04)
SCT Chicks (1996-2006; $N = 4,207$)									
heurak	0.20	0.27	0.11	0.11	0.11	0.06	0.142	0.155	0.179
β	(0.05)	(0.02)	(0.03)	(0.07)	(0.06)	(0.05)	(0.03)	(0.03)	(0.04)

TABLE V

	Coefficients in citation model			Coefficients in predict of R&D effects			R&D rate for papers with at least 1 female author	
	All-female authors	Mixed-gender author other female	Mixed-gender author female	All-female authors	Mixed-gender author other female	Mixed-gender author female	Chen-gross and pub. year	Chen-gross and pub. year
patent destination:							(7)	(8)
out of submission	0.21	0.10	0.12	0.00	0.36	-0.01	0.59	0.94
1990-1999	-0.780 [*]	-0.08	-0.13	(0.06)	0.08	(0.14)	-0.07	—
2000-2009	-0.797 [*]	-0.08	-0.13	(0.06)	0.08	(0.14)	-0.07	—
in years 2010-2012	-0.87	-0.08	-0.08	(0.09)	0.09	(0.15)	0.21	0.12
in years 2013-2017	-0.87	-0.08	-0.08	(0.09)	0.09	(0.15)	0.21	0.12
in years 2018-2017	-0.87	-0.08	-0.08	(0.09)	0.09	(0.15)	0.21	0.12
in years 2018-2017	-0.87	-0.08	-0.08	(0.09)	0.09	(0.15)	0.21	0.12
out of publication	0.18	0.06	0.06	0.04	0.08	-0.03	0.17	0.37
1990-1999	-0.05	0.00	0.00	(0.04)	0.07	(0.10)	-0.07	—
2000-2009	-0.05	0.00	0.00	(0.04)	0.07	(0.10)	-0.07	—
in years 2010-2012	-0.05	0.00	0.00	(0.04)	0.07	(0.10)	-0.07	—
in years 2013-2017	-0.05	0.00	0.00	(0.04)	0.07	(0.10)	-0.07	—
in years 2018-2017	-0.05	0.00	0.00	(0.04)	0.07	(0.10)	-0.07	—

TABLE V
(CONTINUED)

[illegible]

Table V: robustness and heterogeneity

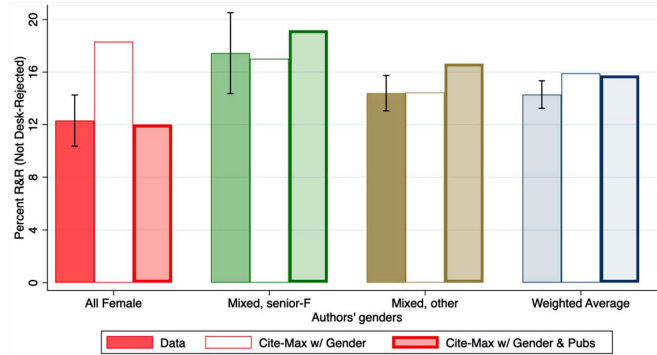


FIGURE V

Implication of Findings for R&R Rate for Female-Authored Papers

This figure presents the implications of the results in [Table IV](#) for the actual and counterfactual R&R rate for teams of authors. The figure breaks down papers by all-female authors, mixed-gender teams with a senior female author, and other mixed-gender teams. Within each group, the first bar plots the observed R&R rate for that group, conditional on a paper not being desk rejected. The second bar reports the counterfactual R&R rate for that category of papers that we would expect to observe if the editorial process aimed to put the weight on the author gender mix variables associated with citation maximization (the details are in the text), keeping the R&R rate constant in each journal-year cell. The third bar reports the counterfactual R&R rate if the editorial process put the citation-maximizing weight not only on the author gender mix variables, but also on the author post-publication variables. The final set of columns presents an average

6.7 Table VI and Figure VI: published papers and referee informativeness

Read:

- Table VI studies accepted and published papers and compares citations across several data sources.
- Published female-authored papers continue to exhibit positive citation patterns in several specifications.
- Figure VI compares the informativeness of male and female referees by referee publication record.
- The relationship between recommendations, citations, and R&R rates is similar across referee gender.

Economic message: Female and male referees appear similarly informative, and female-authored papers continue to look strong in post-publication impact measures.

CITATIONS FOR ACCEPTED PAPERS AND PUBLISHED PAPERS						
set: le of papers: of GS citations measured in:	Editorial Express Accepted papers 2015	Published papers from EconLit Publications in our 4 journals, 2008-2015 2019				
	(1)	(2)	(3)	(4)	(5)	(6)
rs' genders	0.24	0.12	0.12	0.08	0.10	0.09
female	(0.13)	(0.11)	(0.13)	(0.13)	(0.13)	(0.13)
ed-gender author team	-0.11	0.24	0.20	0.13	0.15	0.13
major author female	(0.16)	(0.13)	(0.16)	(0.15)	(0.15)	(0.15)
ed, other	0.09	-0.00	0.04	0.02	0.03	0.03
determined	(0.11)	(0.07)	(0.08)	(0.08)	(0.08)	(0.08)
	-0.40	-0.63	-0.75	-0.75	-0.75	-0.75
	(0.19)	(0.29)	(0.31)	(0.32)	(0.34)	(0.33)
(sum of counts of words) orem, proposition, etc. i words)				-0.12	-0.07	-0.05
irical, table, etc. i words)				(0.04)	(0.04)	(0.04)
l words)				0.20	0.15	0.17
atural, simulation, etc. i words)				(0.04)	(0.04)	(0.04)
laboratory, etc. i words)				0.00	-0.01	-0.01
g of paper content e of model -1 scale)				0.04	0.03	0.00
e of empirical analysis -1 scale)				(0.02)	(0.03)	(0.03)
e of policy -1 scale)					-0.15	-0.12
					(0.15)	(0.15)
					0.41	0.15
					(0.16)	(0.17)
					0.20	0.20
					(0.12)	(0.12)

THE QUARTERLY JOURNAL OF ECONOMICS

TABLE VI (CONTINUED)						
set: le of papers: of GS citations measured in:	Editorial Express Accepted papers 2015	Published papers from EconLit Publications in our 4 journals, 2008-2015 2019				
	(1)	(2)	(3)	(4)	(5)	(6)
positions/theorems < 75th percentile					-0.24	-0.20
mong positives					(0.11)	(0.11)
positions/theorems > 75th percentile					-0.41	-0.29
mong positives					(0.12)	(0.12)
odeling equations < 75th percentile					0.04	0.04
mong positives					(0.08)	(0.08)
odeling equations > 75th percentile					0.24	0.25
mong positives					(0.14)	(0.14)
stimulating equations < 75th percentile					0.07	0.02
mong positives					(0.08)	(0.08)
stimulating equations > 75th percentile					0.01	-0.01
mong positives					(0.09)	(0.09)
ols for fraction referee recs	Yes	No	No	No	No	No
ols for author pub. no. of authors,	Yes	No	Yes	Yes	No	Yes
1 gender-field, and journal-year	No	No	Yes	Yes	Yes	Yes
ols for 2-digit JEL	1,713	1,719	1,719	1,719	1,719	1,719
	0.30	0.28	0.54	0.56	0.56	0.57

a. The sample in column (1) is accepted papers in our main sample. The sample in columns (2)-(7) is published articles from EconLit, in the four journals in our sample in 2015, corresponding approximately to submissions in our sample. For this data set, we measure Google Scholar citations in March 2019 (as opposed to mid-2015 for column (1)).
sum of word counts, we create four groups of words out of 29 words. The qualitative coding of the role of model, empirical analysis, and policy is on a 0-10 scale, which we use to a 0-1 scale. In addition, we counted the number of propositions and theorems, model equations, and estimating equations. For each of these three categories, we indicate paper has a positive number) whether the paper exceeds the 75th percentile for that variable. In column (7), we perform a lasso variable selection, forcing the inclusion of the gender variables.

ARE REFERRERS AND EDITORS GENDER NEUTRAL?

Table VI, part 1

318

THE QUARTERLY JOURNAL OF ECONOMICS

Table VI, continuation

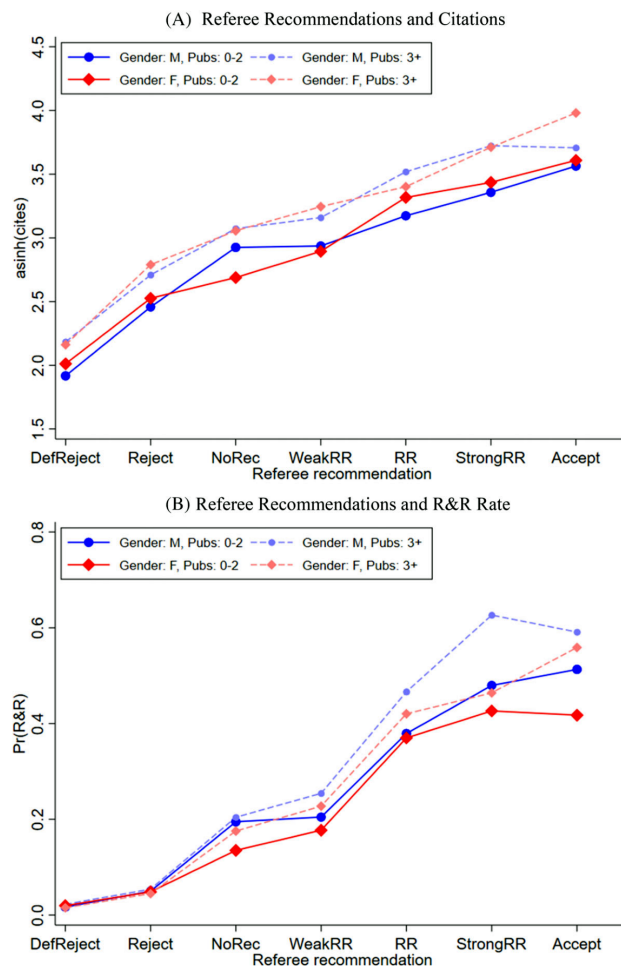


FIGURE VI

Referee Informativeness by Referee Gender and Publications

Panel A shows the weighted $\text{asinh}(\text{citations})$ for a paper receiving a given rec-

6.8 Table VII and Table VIII: desk rejection and editorial delays

Read:

- Table VII shows that editors treat female authors more favorably at the desk-rejection stage than in later R&R decisions.

- Yet even desk rejection still implies a higher bar than a citation-maximizing benchmark would suggest.
- Table VIII examines editorial delay outcomes: days from submission to report, report to decision, rounds for R&R papers, and total time to acceptance.
- The paper finds no systematic gender differences in delay.

Economic message: The higher-bar result is not driven by slower editorial treatment. The main discrepancy lies in the conversion from expected citation value to editorial thresholds.

TABLE VII
DESK REJECTION, IMPACT OF AUTHOR TEAM GENDER

Specification: Dependent variable:	OLS reg. <i>asinh</i> of citations	Probit Indicator for paper not desk rejected	OLS reg. <i>asinh</i> of citations	
	(1)	(2)	(3)	(4)
Authors' genders				
All female	0.24 (0.04)	0.13 (0.04)	0.17 (0.05)	0.15 (0.04)
Mixed-gender author team	0.02 (0.05)	-0.03 (0.09)	0.10 (0.05)	0.15 (0.05)
senior author female	-0.02 (0.03)	-0.03 (0.04)	0.12 (0.04)	0.15 (0.03)
Mixed, other	-0.50 (0.04)	-0.39 (0.05)	-0.41 (0.06)	-0.34 (0.05)
Undetermined				
Author publications in 35 high-impact journals				
Publications: 1	0.53 (0.05)	0.40 (0.04)	0.56 (0.06)	0.55 (0.05)
Publications: 2	0.83 (0.07)	0.62 (0.04)	0.90 (0.09)	0.87 (0.07)
Publications: 3	0.98 (0.08)	0.81 (0.06)	1.06 (0.10)	1.01 (0.09)
Publications: 4-5	1.28 (0.10)	1.05 (0.08)	1.41 (0.14)	1.32 (0.11)
Publications: 6+	1.58 (0.12)	1.34 (0.10)	1.67 (0.16)	1.62 (0.13)
Gender-field controls				
Share female in subfields	0.25 (0.24)	0.21 (0.17)		0.93 (0.21)
Fraction of gender-topic subfields	0.01 (0.06)	0.01 (0.06)		-0.06 (0.07)
NDR indicator	0.41 (0.29)		0.86 (0.07)	0.86 (0.06)
Control function for selection into NDR (value added of the editor)	0.27 (0.15)			
Editor leave-out-mean NDR rate		2.78 (0.36)		
Controls for no. of authors and field	Yes	Yes	No	No
Indicators for journal-year	Yes	Yes	No	Yes
Control for cubic in $P(NDR)$	No	No	Yes	Yes
Number of observations	29,890	29,890	29,890	29,890
R^2 / pseudo R^2	0.28	0.24	0.20	0.27

Notes. Dependent variable for OLS model in columns (1) and (3)-(4) is *asinh* of Google Scholar citations. Dependent variable in probit model in column (2) is indicator for avoiding desk rejection. The control function for selection in column (1) is calculated using predicted probabilities based on column (2). In columns (3) and (4) we control for a cubic polynomial in the probability of non-desk-rejection, built using the specification in column (2). Standard errors clustered by editor are in parentheses.

Table VII: desk rejection

TABLE VIII
REFeree DELAYS, DECISION TIME, AND DURATION OF REVISIONS

	OLS regression of number of days from submission to referee report		Number of days		No. of rounds (for R&Rs)	Total days from first sub. to accept
	(1)	(2)	Sub. to last report received	Submission to editor decision		
yes' genders	-1.36 (0.97)		-1.10 (1.40)	-1.85 (1.49)	0.03 (0.09)	3.59 (30.47)
female	-0.24 (1.83)		0.60 (2.03)	0.71 (3.18)	-0.06 (0.08)	-8.39 (31.80)
oed-gender author team						
senior author female						
oed-gender, other	0.17 (0.72)		1.18 (0.84)	1.82 (1.60)	0.01 (0.05)	19.41 (22.27)
determined gender mix	-1.81 (1.45)		-0.32 (1.97)	-0.71 (2.50)	0.08 (0.11)	38.45 (52.18)
es-gender (omitted: male referee)						
male referee		0.44 (0.98)				
er interactions						
female auth. × Female ref.		1.32 (2.52)				
oed auth. (senior-F) × Female ref.		-2.91 (3.39)				
oed auth. (other) × Female ref.		1.11 (1.92)				
determined auth. × Female ref.		0.87 (4.71)				

Table VIII, part 1

AGE REFERENCE AND EDITORS GENDER NEUTRAL

321

TABLE VIII
(CONTINUED)

	OLS regression of		Number of days		No. of rounds (for R&Rs)	Total days from first sub. to accept
	number of days from submission to referee report		Sub. to last report received	Submission to editor decision		
indicator	(1)	(2)	(3)	(4)	(5)	(6)
			1.71 (1.62)	31.67 (9.26)		
role	Non-desk-rejected papers		Non-desk-rejected papers		R&R papers that are ultimately accepted	
fixed effects	No	Yes	No	No	No	No
controls for referee publications	No	Yes	No	No	No	No
controls for referee recommendations	No	No	Yes	Yes	Yes	Yes
controls for author pub., no. of authors, d & gender-field, journal-year FE	Yes	—	Yes	Yes	Yes	Yes
fixed effects	No	—	Yes	Yes	Yes	Yes
of dependent variable	73.6	73.6	94.9	116.9	2.79	634.1
	38,825	38,825	15,147	15,147	1,713	1,667
	0.31	0.71	0.43	0.44	0.39	0.40

18. The sample in columns (1) and (2) is referee-paper observations, including any referee who returned a review for that paper. Report time is calculated as the number of days from paper submission to referee report submission, rounded to the nearest 10. The sample in columns (3)–(6) is papers observations, including only R&R papers that are ultimately accepted in columns (5)–(6). Decision time is calculated as the number of days from paper submission to referee report submission, rounded to the nearest 10. Clustered standard errors by editor are in parentheses.

7 Short summary

- The paper studies gender in peer review using submissions to four top economics journals.
- Editors assign female referees more often to female-authored papers.
- Male and female referees do not meaningfully differ in how they evaluate female-authored papers.
- Conditional on referee recommendations and controls, female-authored papers receive about 25% more citations than similar male-authored papers.
- This implies female-authored papers face a higher implicit standard in the review process.
- Editors largely follow referees, so R&R decisions under-admit female-authored papers relative to a citation-maximizing benchmark.
- Desk rejection decisions are more favorable to female authors, but still not fully citation-maximizing.
- The paper finds no gender difference in referee informativeness or editorial delay.

8 Conclusion

8.1 Core conclusion

Card, DellaVigna, Funk, and Iriberry show that the main gender issue in economics journal peer review is not differential bias by male versus female referees. Instead, female-authored papers appear to receive similar recommendations and editorial treatment despite higher subsequent citations, indicating a higher bar for female authors.

8.2 Economic intuition

A referee recommendation is an imperfect signal of paper quality. If female-authored papers receiving the same recommendations as male-authored papers later receive more citations, then the signal is being interpreted in a way that underweights female-authored papers. Editors' strong reliance on referees transmits this higher bar into R&R decisions.

8.3 Takeaway

The paper provides an empirical audit of a central gatekeeping institution in economics. It shows that gender neutrality cannot be assessed only by comparing male and female referees or raw acceptance rates. A better standard is whether editorial decisions are aligned with later impact across author groups. By that standard, the process appears to impose a higher threshold on female-authored papers.